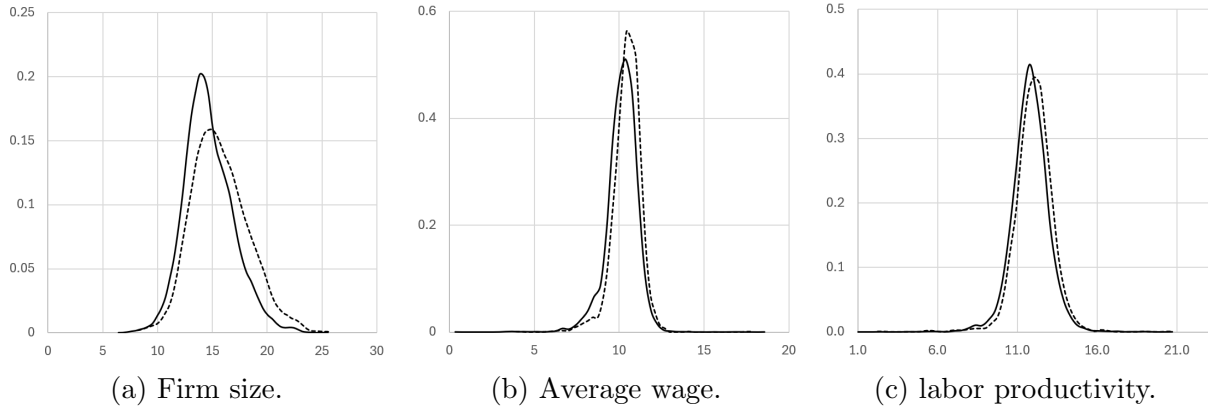


that is plausibly exogenous to the EU context. This cross-country comparison allows us to construct an instrumental variable or control function that reflects external drivers of AI adoption, thereby improving the causal interpretation of our estimates.

Last but not least, we explore the link between AI adoption and productivity in a stylized distributional exercise. Figure 5 compares the distribution of firm size, average wage per employee and labor productivity for firms that adopted AI and those that did not. The distributions for firms using AI are shifted to the right. Firms that have adopted big data and AI technologies are, on average, larger (in line with descriptive evidence discussed above), pay higher wages and have slightly higher productivity. This provides preliminary evidence that firms that use AI tend to perform better than other companies.

Figure 5: Distributions of firm characteristics for firms using AI (dashed) and firms not using AI (solid line)



Note: The graphs show the distributions net of country and sector fixed effects. Firm size is measured using the logarithm of total assets. EU firms only. *Source:* EIBIS 2019-2024.

4 Empirical methodology

We estimate the causal impact of AI adoption on firm-level productivity with a two-stage empirical strategy using EIBIS data. Our primary specification is a linear probability model, which allows for a straightforward interpretation of the marginal effects of AI adoption on the likelihood of observing productivity improvements. The identification challenge is that AI adoption is potentially endogenous, i.e. firms that are more productive or better managed may be more likely to adopt AI technologies, leading to biased

estimates if not properly addressed.

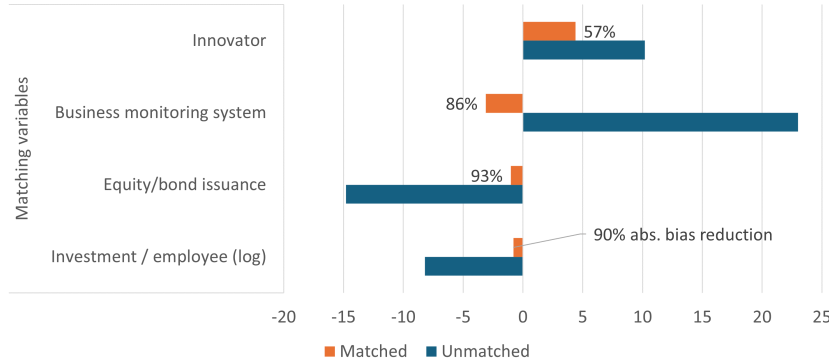
To mitigate endogeneity concerns, we develop an identification strategy inspired by [Rajan and Zingales \(1998\)](#). Their seminal insight is that characteristics measured in a benchmark economy can serve as an exogenous source of variation when applied to other countries. Whereas [Rajan and Zingales \(1998\)](#) assign US sectoral rankings of financial dependence to all countries, we extend this logic using a more granular, firm-level design. Specifically, we construct an external proxy for AI exposure by matching each EU firm to a set of comparable US firms – similar in sector, size class, investment intensity, innovation activity, financing structure and management practices – and transferring the US firms’ AI adoption scores to the EU firm. Because US firms operate in a technological and institutional environment that is arguably exogenous to the EU, the matched US adoption measure captures the component of AI exposure that is unrelated to EU-specific policies, economic conditions or unobserved firm characteristics. This firm-level matching approach, which exploits cross-continental variation while tightly conditioning on underlying fundamentals, refines earlier sector-level strategies, such as the use of US sectoral AI adoption rates as in [Gambacorta et al. \(2025\)](#).

The matching between EU and US firms is implemented using a stratified propensity score matching (PSM) approach, which ensures comparability across firms based on observable characteristics. Building on the firm characteristics reported in Section 3.1, our strategy is the following. We stratify firms by two main categorical dimensions that seem to be crucial for AI adoption: sector and size class. Within each stratum, we match firms on variables for which the observed difference is the highest between AI and non-AI firms, as reported in Table 1. The chosen four key covariates are: (i) investment intensity per employee (in log), (ii) a binary indicator for whether a company issued bonds or equity, (iii) a binary indicator for innovation activity, and (iv) a binary indicator for the use of a strategic business monitoring system. The variables are not only motivated by our initial investigation of the main drivers of AI adoption in the previous Section but are also closely linked to the literature on firm productivity and technology adoption, highlights the importance of financial capacity, innovation orientation and managerial quality as determinants of both AI adoption and productivity outcomes ([Bloom, Sadun, and Van Reenen, 2016](#); [Hall, 2010](#)).⁴

⁴We confirm that the selected variables have high predictive power over AI adoption decisions in the EU and US, using a stylized linear regression model. The results are presented in Table B.1 in the Appendix.

The use of stratified PSM helps us to address selection bias by ensuring that matched firms are similar in terms of observable characteristics that jointly influence both AI adoption and productivity. Moreover, by using US firms as a benchmark, we exploit variation in AI adoption that is plausibly exogenous to EU-specific institutional or policy environments. This strategy enhances the credibility of our identification and allows us to interpret the estimated effects as causal under the assumption of conditional independence (Rosenbaum and Rubin, 1983; Dehejia and Wahba, 2002).⁵

Figure 6: Balancing properties before and after matching



Note: The bars present the standardized percentage biases before and after matching for the selected variables, with the numbers highlighting the absolute bias reduction after matching (positive values mean that EU firms score higher). For each EU firm, the matched sample includes the five nearest-neighbour US firms based on propensity scores calculated using the specified matching variables. *Source:* EIBIS 2019-2024.

Overall, our matching procedure is effective, as evidenced by the substantial reduction in covariate imbalances between EU and US firms. Prior to matching, US firms exhibit a significantly higher investment intensity – by approximately 8.2 percent – compared to their EU counterparts. After matching, this difference narrows to just 0.8 percent, representing a 90% reduction in the initial gap (Figure 6). Similar improvements are observed across all other matching variables, indicating that the matched US firms closely

⁵We use the k -nearest neighbour approach to find the best possible match between US and EU firms. We choose $k = 5$ to balance the trade-off between having enough variation for a reasonable match while reducing potential overlap across matches. Lower values of k risk putting too much weight on any given US company. In turn, since our sample of US firms is considerably smaller than that of EU firms, expanding the set of nearest neighbours used for matching risks a high degree of overlap in the matching for any given pair of EU firms. Nonetheless, our main results remain robust to matching procedures with $k = 3$ and $k = 7$.

resemble EU firms in terms of observable characteristics.⁶

We run pooled cross-section regressions to investigate the effects of AI adoption on labor productivity and employment :

$$y_{it} = \beta_1 \text{Adopter}_{it} + \beta_2 \mathbf{X}_{it-1} + \nu_{age} + \phi_{size} + \mu_{cst} + \epsilon_{it}, \quad (1)$$

where i denotes the firm and t the year. The dependent variable y_{it} is either labor productivity or employment at the firm level. The explanatory variable Adopter_{it} takes value 1 for AI-adopting firms and 0 otherwise (either the actual series or the AI adoption instrumented after matching with US firms, as discussed above)⁷ and $\mathbf{X}_{it-1}$ is a vector of controls which includes investment, profitability, financial leverage and the logarithm of total assets (all lagged by one period to moderate simultaneity effects). We add a full set of fixed effects, including (i) size groups $\phi_{size} \in \{\text{micro}, \text{small}, \text{medium}, \text{large}\}$ to control for the possibility that AI adoption differs for different size categories, (ii) age groups $\nu_{age} \in \{< 5y, 5 - 10y, 10 - 20y, > 20y\}$ to account for potentially confounding variation that may be a function of firm age, and (iii) country-sector-wave fixed effects to absorb possible idiosyncratic shocks in AI adoption rates across firms in the same country-sector-time group. We control for the three sets of fixed effects in all specifications.⁸

To investigate potential heterogeneity in the effects of AI adoption on labor productivity, we augment Equation (1) by introducing interaction terms between AI adoption and two key dimensions: (i) firm size group and (ii) the share of investment allocated to different categories of tangible and intangible assets. For the latter, we use EIBIS reported data on investment in the last financial year (divided by total assets) attributed to any of the six categories: (i) land and buildings, (ii) machinery and equipment, (iii) research and development, (iv) data, software and IT, (v) training and (vi) business processes and organization.

When estimating regressions, it is important to account for the fact that we use

⁶The effectiveness of the matching procedure is further supported by the strong explanatory power of matching covariates in accounting for differences in AI adoption rates between US-matched and non-matched firms, as documented in the Appendix.

⁷Since the matching algorithm uses k -nearest neighbors, a US-matched value is set to 1 if the majority of the nearest neighbors are AI adopters, and 0 otherwise.

⁸Even in the presence of an exogenous instrument, there may still be unobserved global shocks or sectoral trends affecting both US and EU firms. We account for this by running the baseline model with different set of fixed effects in Table B.2 in the Appendix. In particular, we separate country-wave from sector-wave fixed effects.